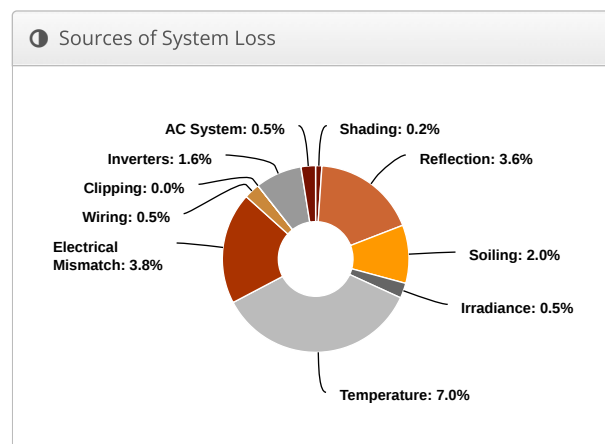
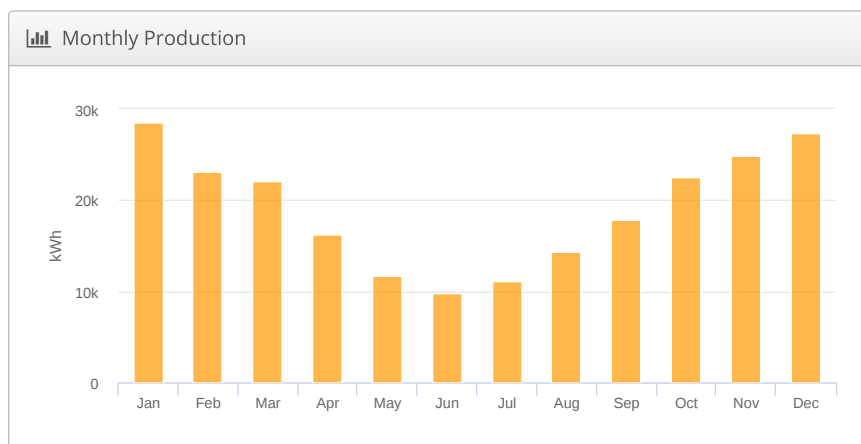
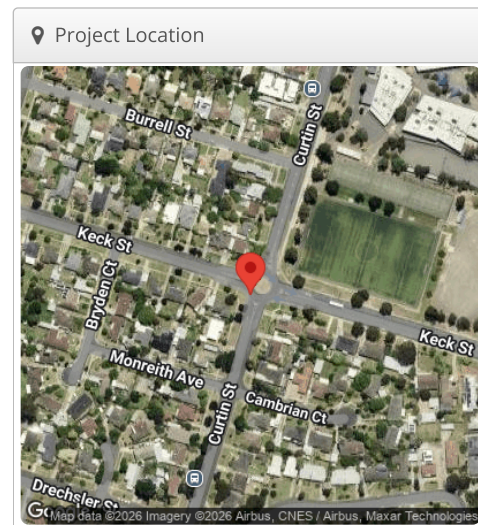


Design 1 LTU Carpark BEndigo, Flora Hill VIC 3550

Report	
Project Name	LTU Carpark BEndigo
Project Address	Flora Hill VIC 3550
Prepared By	Rana Mitra rana@enhar.com.au
	

System Metrics	
Design	Design 1
Module DC Nameplate	159.08 kW
Inverter AC Nameplate	150.00 kW Load Ratio: 1.06
Annual Production	229.8 MWh
Performance Ratio	81.7%
kWh/kWp	1,444.8
Weather Dataset	TMY, 10km Grid, Meteonorm 8 (meteonorm_v8)
Simulator Version	1a99afbe8f-c1102635c5-2a145ef897-e287566cff



⚡ Annual Production			
	Description	Output	% Delta
Irradiance (kWh/m²)	Annual Global Horizontal Irradiance	1,644.9	
	POA Irradiance	1,768.4	7.5%
	Shaded Irradiance	1,764.6	-0.2%
	Irradiance after Reflection	1,701.5	-3.6%
	Irradiance after Soiling	1,667.5	-2.0%
	Total Collector Irradiance	1,667.5	0.0%
Energy (kWh)	Nameplate	265,358.2	
	Output at Irradiance Levels	263,991.0	-0.5%
	Output at Cell Temperature Derate	245,445.1	-7.0%
	Output after Electrical Mismatch	236,026.0	-3.8%
	Optimal DC Output	234,748.6	-0.5%
	Constrained DC Output	234,743.4	0.0%
	Inverter Output	230,987.5	-1.6%
	Energy to Grid	229,832.6	-0.5%
Temperature Metrics			
Avg. Operating Ambient Temp		17.1 °C	
Avg. Operating Cell Temp		35.4 °C	
Simulation Metrics			
Operating Hours		4634	
Solved Hours		4634	

☁ Condition Set												
Description	Condition Set 1											
Weather Dataset	TMY, 10km Grid, Meteonorm 8 (meteonorm_v8)											
Solar Angle Location	Meteo Lat/Lng											
Transposition Model	Perez Model											
Temperature Model	Sandia Model											
Temperature Model Parameters	Rack Type				a		b		Temperature Delta			
	Fixed Tilt				-3.56		-0.075		3°C			
	Flush Mount				-2.81		-0.0455		0°C			
	East-West				-3.56		-0.075		3°C			
	Carport				-3.56		-0.075		3°C			
Soiling (%)	J	F	M	A	M	J	J	A	S	O	N	D
	2	2	2	2	2	2	2	2	2	2	2	2
Albedo	J	F	M	A	M	J	J	A	S	O	N	D
	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20
Rear Mismatch Loss	10%				Rear Shading Factor				5%			
Module Transparency	0%											
Irradiation Variance	5%											
Cell Temperature Spread	4° C											
Module Binning Range	-2.5% to 2.5%											
AC System Derate	0.50%											
Module & Component Characterizations	Type	Component					Characterization					Bifacial
	Module	LR7-54HVD-485M (Longi)					Spec Sheet Characterization, PAN					True
	Inverter	SG50CX (Sungrow)					Spec Sheet					N/A

Components		
Component	Name	Count
Inverters	SG50CX (Sungrow)	3 (150.00 kW)
Strings	10 AWG (Copper)	15 (608.3 m)
Module	Longi, LR7-54HVD-485M (485W)	328 (159.08 kW)

Wiring Zones			
Description	Combiner Poles	String Size	Stringing Strategy
Wiring Zone	-	17-25	Along Racking

Field Segments									
Description	Racking	Orientation	Tilt	Azimuth	Intrarow Spacing	Frame Size	Frames	Modules	Power
Field Segment 1	Flush Mount	Landscape (Horizontal)	10°	358.46567°	0.0 m	2x	N/A	328	159.08 kW

Detailed Layout2

